

TA-1630

AEP Model
UK Model
Canadian Model
E Model



INTEGRATED STEREO AMPLIFIER

SPECIFICATIONS

GENERAL

Power Requirements: 110, 127, 220 or 240 V ac adjustable,
 50/60 Hz (AEP, UK, E Model)
 120 V ac, 60 Hz (Canadian Model)

Power Consumption: 210 W (AEP, E Model)
 150 W (UK Model)
 100 W (Canadian Model)

AC Outlets (Canadian Model): 1 switched 200 W
 2 unswitched 200 W

Dimensions: Approx.
 388 (w) x 145 (h) x 286 (d) mm
 15 1/8 (w) x 5 3/4 (h) x 11 1/4 (d) inches
 including projecting parts and controls

Weight: Approx. 6.7 kg, 14 lb 12 oz (net)
 Approx. 8.5 kg, 18 lb 12 oz (with shipping
 carton)

Power Bandwidth: 10 – 40,000 Hz, IHF

Damping Factor: 40 (8Ω, at 1 kHz)

Harmonic Distortion: Less than 0.5 % at rated output
 Less than 0.1 % at 1 W output

IM Distortion: Less than 0.5 % at rated output
 (60 Hz: 7 kHz = 4:1) Less than 0.1 % at 1 W output

Residual Noise: Less than 0.25 μW (8Ω)

Frequency Response:

PHONO	RIAA equalization curve ±0.5 dB
TUNER	
AUX	
TAPE 1, 2	20 – 100,000 Hz ± 1 dB
REC/PB (input (AEP, UK Model))	

AMPLIFIER SECTION

Continuous RMS

Power Output: Both channels driven simultaneously
 (rated output) At 20 – 20,000 Hz
 22 + 22 W (8Ω)
 (Less than 0.5 %
 harmonic distortion) At 1 kHz
 25 + 25 W (8Ω)

According to DIN 45500
 23 + 23 W (8Ω)

Dynamic Power Output: 120 W (8Ω)
 (IHF constant power
 supply method)

(Continued on page 2.)

SONY®

SERVICE MANUAL

Input:

	Sensitivity	Impedance	Maximum input capability (THD 0.8 %)	S/N (weighting network, input level)
PHONO	2.5 mV (-50 dB)	50 kΩ	120 mV	70 dB (B, 2.5 mV)
TUNER AUX TAPE 1, 2 REC/PB (AEP, UK Model)	200 mV (-12 dB)	100 kΩ	—	90 dB (A, 250 mV)

Output:

	Output level	Impedance
REC OUT 1	200 mV (-12 dB)	10 kΩ
REC/PB (AEP, UK Model)	30 mV (-28 dB)	82 kΩ

- HEADPHONES:** Accepts low and high impedance headphones

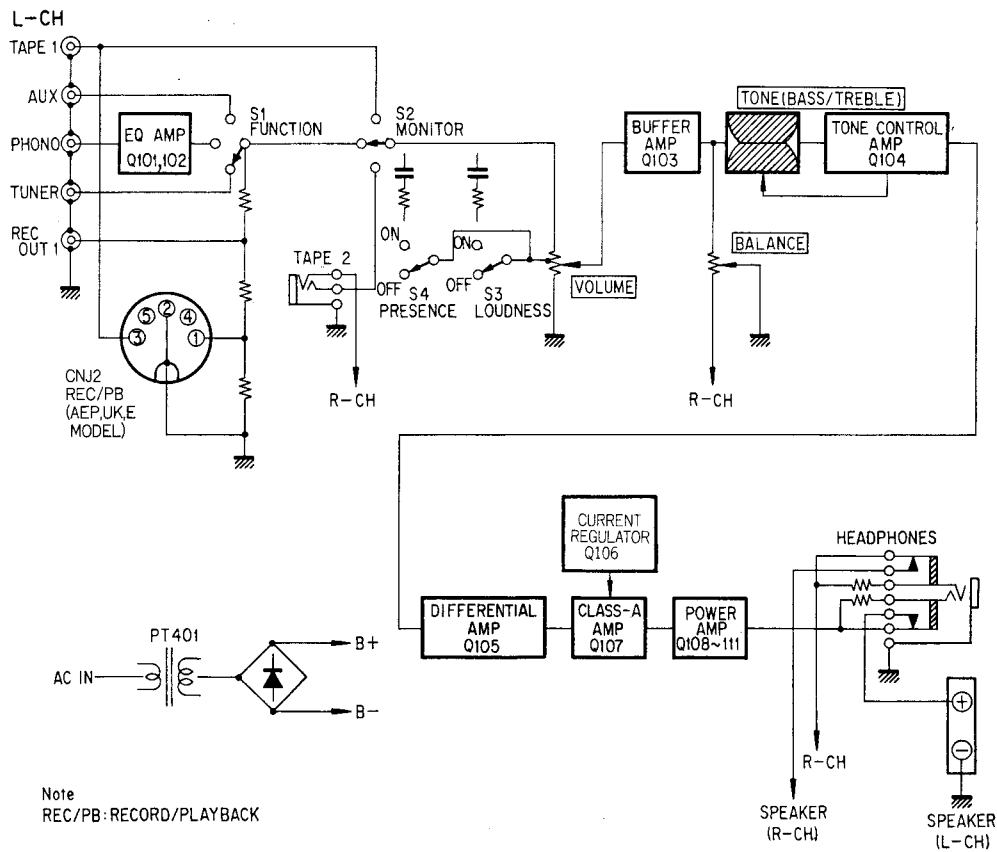
SPEAKER: Accepts 4 – 16 Ω speakers (AEP, Canadian, E Model)
Accepts 8 – 16 Ω speakers (UK Model)

Tone Controls: BASS ±9 dB at 100 Hz
(turnover frequency 250 Hz)
- TREBLE ±6.5 dB at 10 kHz
(turnover frequency 4 kHz)

Loudness Controls: +8.8 dB at 50 Hz, +2.5 dB at 10 kHz
(Attenuation: 30 dB)

Presence: +2.5 dB at 1 kHz (Attenuation: 30 dB)

1. BLOCK DIAGRAM

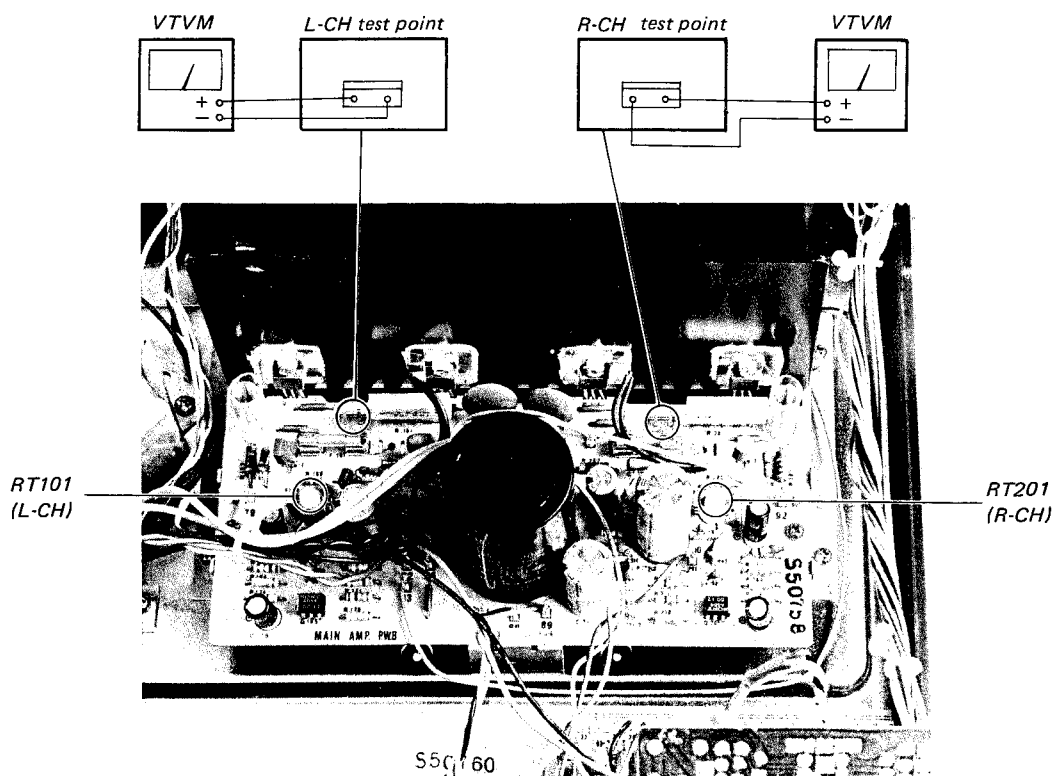


2. IDENTIFICATION OF SET

E, AEP Model	UK Model
SONY® INTEGRATED STEREO AMPLIFIER MODEL NO. TA-1630 AC 110, 127, 220, 240V~50/60Hz 210W SERIAL NO. _____ MADE IN JAPAN	SONY® INTEGRATED STEREO AMPLIFIER MODEL NO. TA-1630 AC 110, 127, 220, 240V~50/60Hz 150W SERIAL NO. _____ MADE IN JAPAN
Canadian Model	
SONY® INTEGRATED STEREO AMPLIFIER MODEL NO. TA-1630 AC 120V 60Hz 100W SERIAL NO. _____ MADE IN JAPAN	

3. DC BIAS ADJUSTMENT

Adjust RT101 (L-CH) and RT201 (R-CH) for 25 mV dc reading on VTVM.



— Conductor Side —

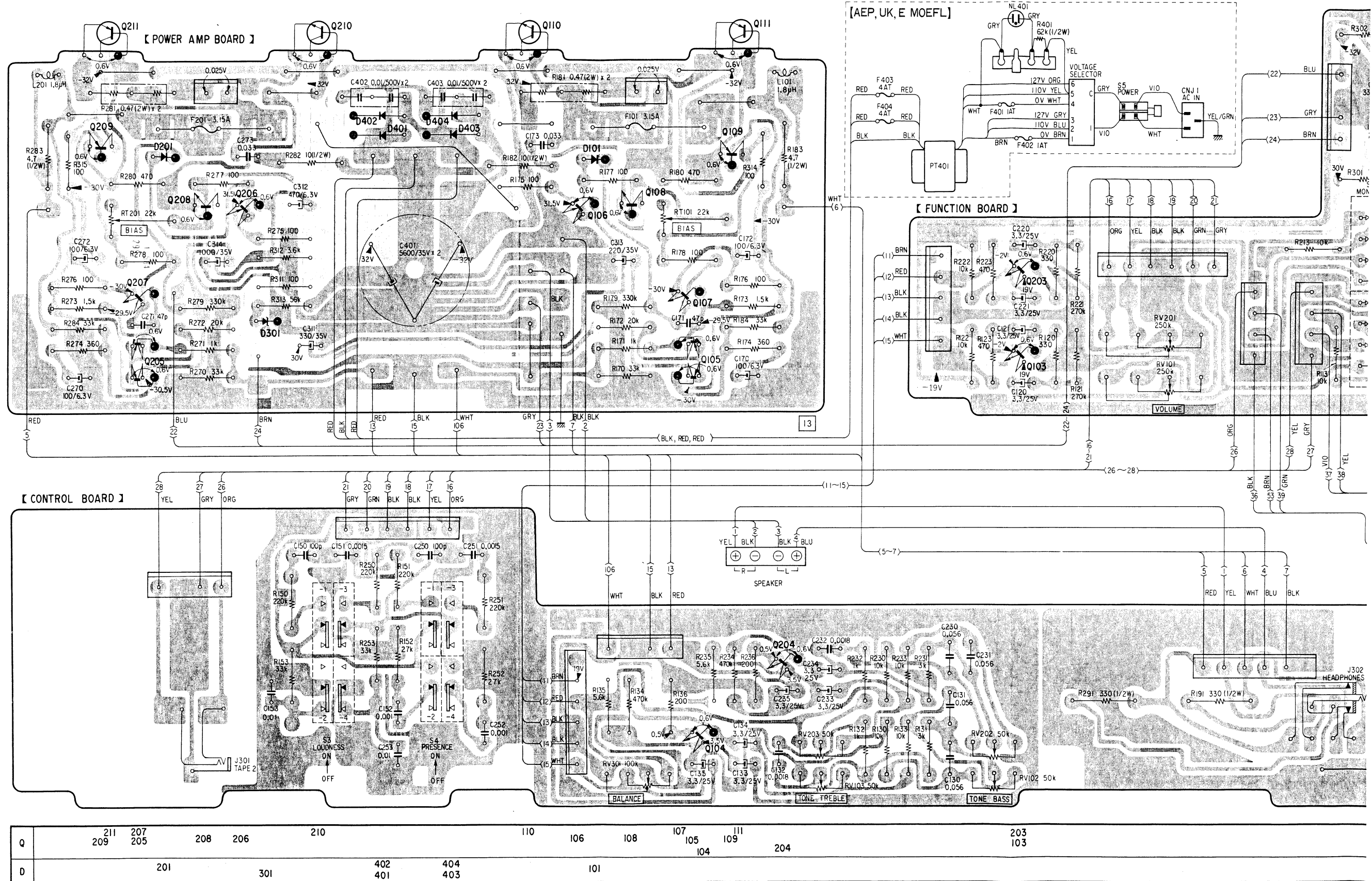
A diagram of a vertical column. The top part is a hexagonal prism, and the bottom part is a circular cylinder. The column is labeled with 'E' on the left, 'C' in the middle, and 'B' on the right at the base. A dashed line runs vertically through the center of the column.

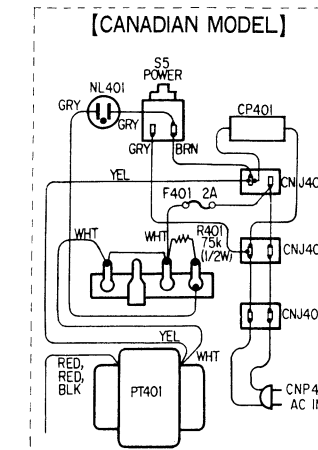
A diagram of a 3-pin transistor package. The package is shown in perspective, with three pins extending downwards. The pins are labeled from left to right as E, B, and C, representing Emitter, Base, and Collector respectively.

anode

cathod

1

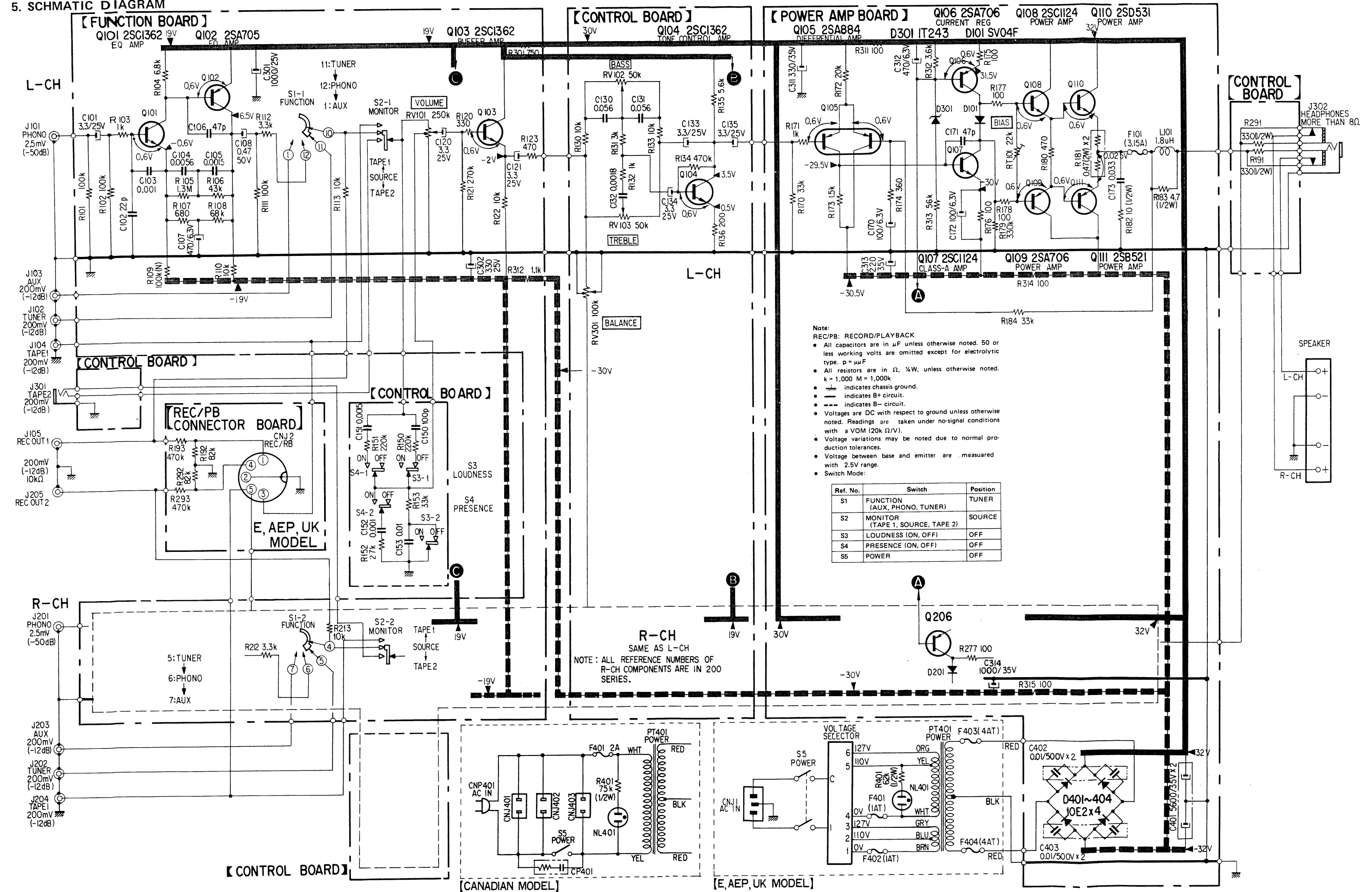




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TA-1630	TA-1630
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5. SCHMATIC DIAGRAM



6. DISASSEMBLY

SIDE BOARD REMOVAL

Remove four screws (± BW 4 x 22).



Two screws are on the opposite side.

FRONT PANEL REMOVAL

1. Pull the knobs out.
2. Remove four screws marked ● (BV 3 x 6 Fig. A and B).

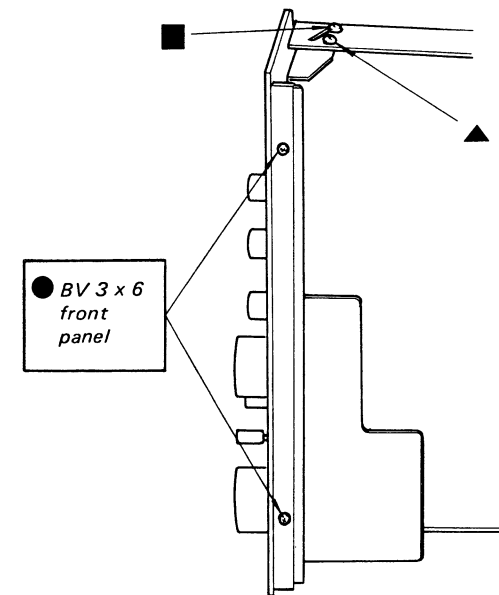


Fig. A Top view

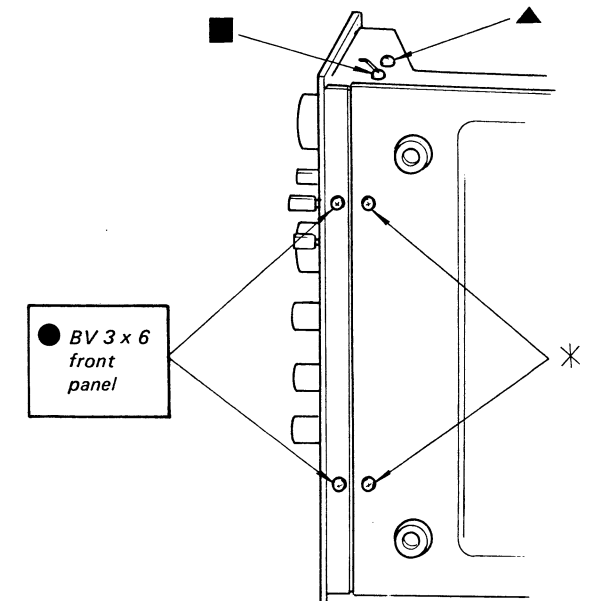
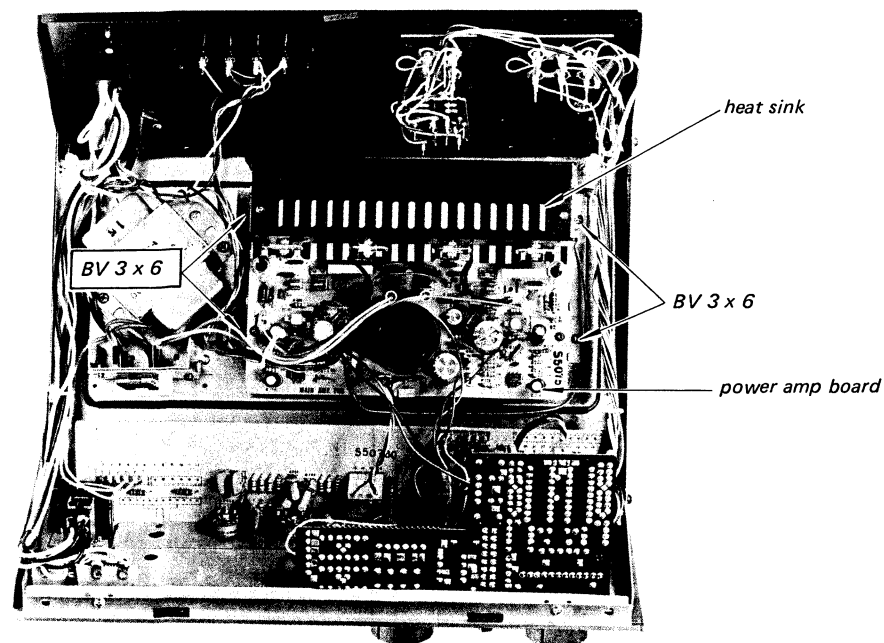


Fig. B Bottom view

POWER AMP BOARD REMOVAL

1. Remove four screws (BV 3 x 6).
2. Remove power amp board along with heat sink.



CONTROL BOARD REMOVAL

1. Remove the front panel.
2. Remove three screws marked ★ (BV 3 x 6, Fig. C).

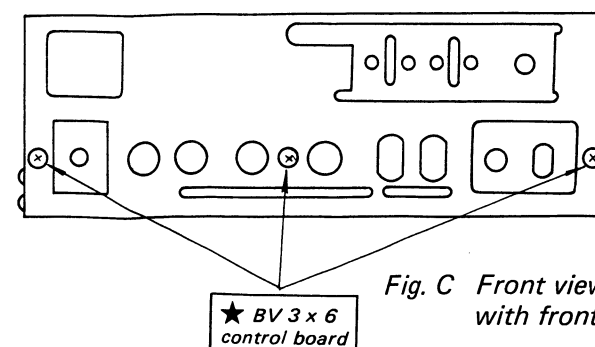


Fig. C Front view with front panel removed

HOW TO RAISE THE CIRCUIT BOARD (FUNCTION and CONTROL BOARD)

1. Remove two screws marked ▲ (BV 3 x 6, Fig. A and B).
2. Loosen two screws marked ■ (BV 3 x 6, Fig. A and B).
3. Remove two screws marked ✕ (BV 3 x 6, Fig. B).
4. Tilt the front panel section in the arrow direction as shown in Fig. D.

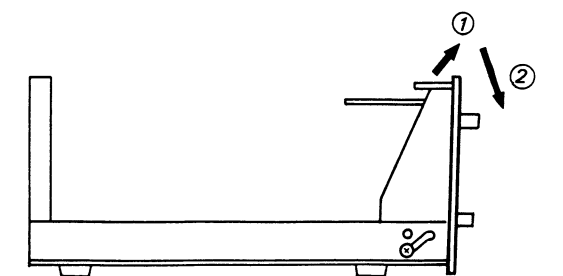
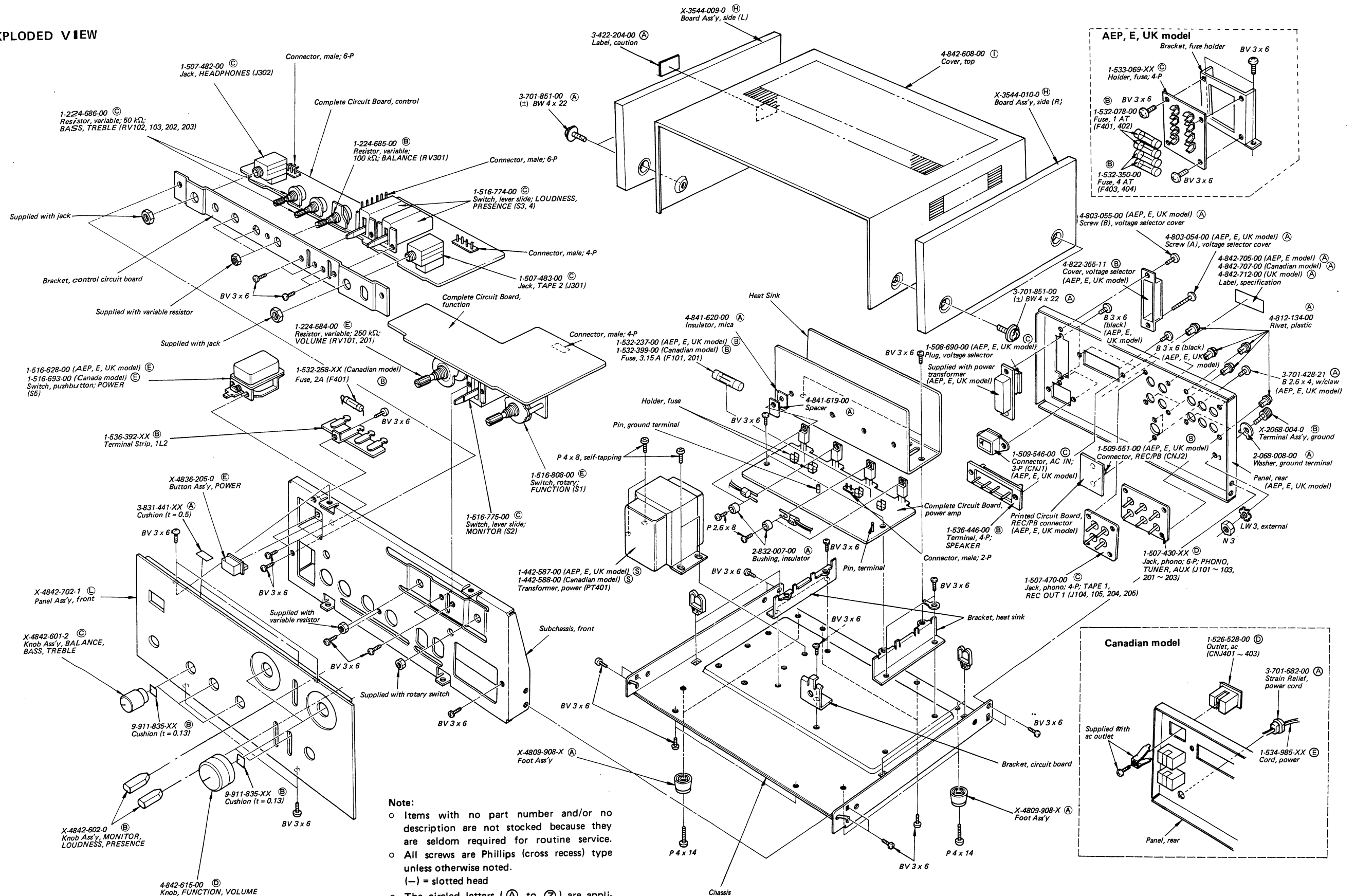


Fig. D Side view

TA-1630 TA-1630

7. EXPLODED VIEW



TA-1630 TA-1630

- o The circled letters (A to Z) are applicable for European model only.
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8. ELECTRICAL PARTS LIST

Ref. No.	Part No.	Description	Ref. No.	Part No.	Description	Ref. No.	Part No.	Description
SEMICONDUCTORS								
Transistors						JACKS		
Q101, 201		B 2SC1362	C150, 250	1-102-973-11	A 100 p ceramic	J101 ~ 103	1-507-430-XX	D Phono, 6-P; PHONO, TUNER, AUX
Q102, 202		B 2SA705	C151, 251	1-108-827-12	A 0.0015 mylar	J201 ~ 203		
Q103, 203		B 2SC1362	C152, 252	1-108-825-12	A 0.001 mylar	J104, 204	1-507-470-00	C Phono, 4-P; TAPE 1, REC OUT 1
Q104, 204			C153, 253	1-108-837-12	A 0.01 mylar	J105, 205		
Q105, 205		D 2SA884	C170, 270	1-121-413-11	E 100 6.3 V	J301	1-507-483-00	C TAPE 2
Q106, 206		D 2SA706	C171, 271	1-101-880-11	A 47 p ceramic	J302	1-507-482-00	C HEADPHONES
Q107, 207		C 2SC1124	C172, 272	1-121-413-11	E 100 6.3 V	FUSES		
Q108, 208		C 2SC1124	C173, 273	1-105-679-12	A 0.033 mylar	F101, 201	1-532-237-00	B 3.15 A (AEP, E, UK model)
Q109, 209		D 2SA706	C301	1-123-066-11	B 1000 25 V		1-532-399-00	B 3.15 A (Canadian model)
Q110, 210		D 2SD531	C302	1-123-065-11	B 330 25 V (explosion proof)	F401	1-532-268-XX	B 2 A (Canadian model)
Q111, 211		E 2SB521	C311	1-123-064-11	B 330 35 V	F401, 402	1-532-078-00	B 1 AT (AEP, E, UK model)
Diodes			C312	1-123-077-11	B 470 6.3 V (explosion proof)	F403, 404	1-532-350-00	B 4 AT (AEP, E, UK model)
D101, 201		C SV04F	C313	1-123-063-11	B 220 35 V	MISCELLANEOUS		
D301		B 1T243	C314	1-121-945-11	C 1000 35 V	CNJ1	1-509-546-00	C Connector, AC IN, 3-P (AEP, E, UK model)
D401 ~ 404		B 10E2	C401	1-125-097-11	H (5600 35 V) x 2	CNJ2	1-509-551-00	B Connector, REC/PB (AEP, E, UK model)
CAPACITORS			C402, 403	1-102-355-11	A (0.01 500 V) x 2 ceramic	CNJ401 ~ 403	1-526-528-00	D Outlet, ac (Canadian model)
All capacitors are in μ F and of electrolytic unless otherwise noted. 50 or less working volts are omitted except for electrolytic type. (p = μ F)			RESISTORS			NL401	1-519-139-11	B Lamp, neon
C101, 201	1-121-913-11	A 3.3 25 V	All resistors are in ohms. Regular-type $\frac{1}{4}$ W carbon resistors are omitted. Check schematic diagram for resistance values. k = 1000			PT401	1-442-587-00	S Transformer, power (AEP, E, UK model)
C102, 202	1-102-959-11	A 22 p ceramic	R181, 281	1-217-359-11	B (0.47 2W) x 2 metal		1-442-588-00	S Transformer, power (Canadian model)
C103, 203	1-108-792-12	A 0.001 mylar	R182, 282	1-202-525-11	A 10 $\frac{1}{2}$ W composition		1-508-690-00	C Plug, voltage selector (AEP, E, UK model)
C104, 204	1-108-801-12	A 0.0056 mylar	R183, 283	1-202-517-11	A 4.7 $\frac{1}{2}$ W composition		1-231-057-31	B Encapsulated Component (Canadian model)
C105, 205	1-108-794-12	A 0.0015 mylar	R191, 291	1-202-561-11	A 330 $\frac{1}{2}$ W composition		1-533-069-XX	C Holder, fuse; 4-P (AEP, E, UK model)
C106, 206	1-101-880-11	A 47 p ceramic	R401	1-202-616-11	A 62 k $\frac{1}{2}$ W composition (AEP, E, UK model)		1-534-985-XX	E Cord, power (Canadian model)
C107, 207	1-123-077-11	B 470 6.3 V (explosion proof)		1-202-618-11	A 75 k $\frac{1}{2}$ W composition (Canadian model)		1-536-392-XX	B Terminal Strip, 1L2
C108, 208	1-121-911-11	A 0.47 50 V	RT101, 201	1-224-646-XX	B 22 k adjustable		1-536-446-00	B Terminal, 4-P; SPEAKER
C120, 220	1-121-913-11	A 3.3 25 V	RV101, 201	1-224-684-00	E 250 k variable, VOLUME	ACCESSORIES		
C121, 221			RV102, 202	1-224-686-00	C 50 k variable, BASS, TREBLE	X-4842-703-3	F Carton Ass'y (AEP, UK, Canadian, E model)	
C130, 230	1-108-846-12	A 0.056 mylar	RV103, 203			1-534-819-00	E Cord, power (UK model)	
C131, 231			RV301	1-224-685-00	B 100 k variable, BALANCE	3-780-750-11	B Manual, instruction (AEP, UK, Canadian, E model)	
C132, 232	1-108-828-12	A 0.0018 mylar	SWITCHES					
C133 ~ 135	1-121-913-11	A 3.3 25 V	S1	1-516-808-00	E Rotary, FUNCTION			
C233 ~ 235			S2	1-516-775-00	C Lever Slide, MONITOR			
			S3, 4	1-516-774-00	C Lever Slide, LOUDNESS, PRESENCE			
				1-516-628-00	E Pushbutton, POWER (AEP, UK model)			
			SS	1-516-693-00	E Pushbutton, POWER (Canadian model)			

HARDWARE NOMENCLATURE

Screw: P 3 x 10
 L: Length in mm
 D: Diameter in mm
 P: Type of head
 Indicated slotted-head only.
 Unless otherwise indicated, it means cross-recessed head (Phillips type).

Nut, Washer, Retaining ring:

N 3
 Diameter of usable screw or shaft
 Reference designation

Reference Designation	Shape	Description	Remarks
SCREWS			
P		pan-head screw	binding-head (B) screw for replacement
PWH		pan-head screw with washer face	binding-head (B) screw and flat washer for replacement
PS PSP		pan-head screw with spring washer	binding-head (B) screw and spring washer for replacement
PSW PSPW		pan-head screw with spring and flat washers	binding-head (B) screw and spring and flat washers for replacement
R		round-head screw	binding-head (B) screw for replacement
K		flat-countersunk-head screw	binding-head (B) screw for replacement
RK		oval-countersunk-head screw	
B		binding-head screw	
T		truss-head screw	
F		flat-fillister-head screw	
RF		fillister-head screw	
BV		brazer-head screw	

Reference Designation	Shape	Description	Remarks
SELF-TAPPING SCREWS			
TA		self-tapping screw	ex: TA, P 3 x 10
PTP		pan-head self-tapping screw	binding-head self-tapping (TA, B) screw for replacement
PTPWH		pan-head self-tapping screw with washer face	binding-head self-tapping (TA, B) screw and flat washer for replacement
PTTWH		pan-head thread-rolling screw with washer face	binding-head (B) screw and flat washer for replacement
SET SCREWS			
SC		set screw	
SC		hexagon-socket set screw	ex: SC 2.6 x 4, hexagon socket
NUT			
N		nut	
WASHERS			
W		flat washer	
SW		spring washer	
LW		internal-tooth lock washer	ex: LW3, internal
LW		external-tooth lock washer	ex: LW3, external
RETAINING RINGS			
E		retaining ring	
G		grip-type retaining ring	